SPIKING MEETS CAUSALITY: EFFICIENT GRANGER CAUSAL DISCOVERY WITH SPIKING NEURAL NETWORKS

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Motivations

➢ **The Limitation of Traditional Granger Causal Discovery (GCD)**

Traditional VAR models are inherently linear. They struggle to capture complex, nonlinear dependencies in real-world systems, severely limiting their accuracy and scalability.

➢ **The Scalability Bottleneck of Neural GCD**

Deep learning approaches address nonlinearities but incur steep computational costs. Their dense, continuous-value operations result in high MACs and heavy power consumption, making large-scale causal discovery challenging.

➢ **SpikGC (Ours)**

While SNNs offer event-driven and ultra-low-power computation , their capability for causal inference remains largely unexplored. SpikGC addresses this gap, achieving an optimal balance between nonlinear modeling power and remarkable efficiency.

Contributions

➢ **Pioneering SNN-based Granger Causal Discovery**

Pioneers the integration of Granger causality with Spiking Neural Networks. Leverages event-driven dynamics to model complex, nonlinear temporal dependencies while dramatically slashing energy consumption.

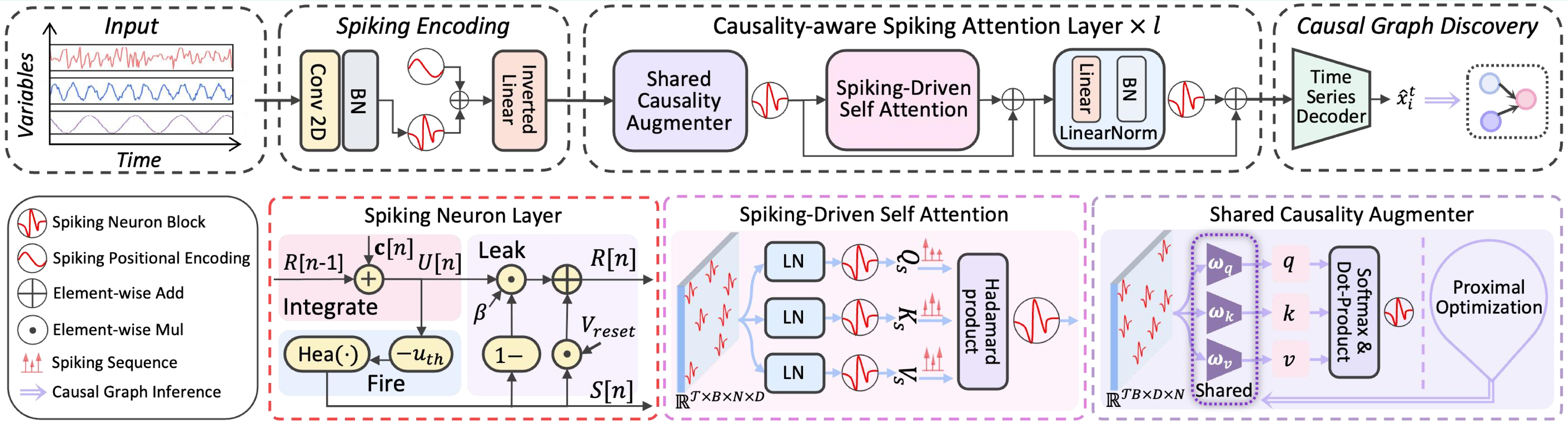
➢ **Novel Causality-Aware Spiking Attention**

Introduces a novel, ultra-low-power attention mechanism. It features (1) a Shared Causality Augmenter to jointly extract variable-wise causal structures, and (2) a Spiking-Driven Self-Attention that achieves $O(ND)$ linear complexity via Hadamard-product masking—completely eliminating energy-heavy softmax and scaling operations.

➢ **SOTA Performance and Superior Efficiency**

Achieves SOTA performance, substantially surpassing existing neural methods (e.g., NGC, JRNGC, CUTS) in both precision and computational efficiency.

Methods



Results

Cuasal Discovery Performance

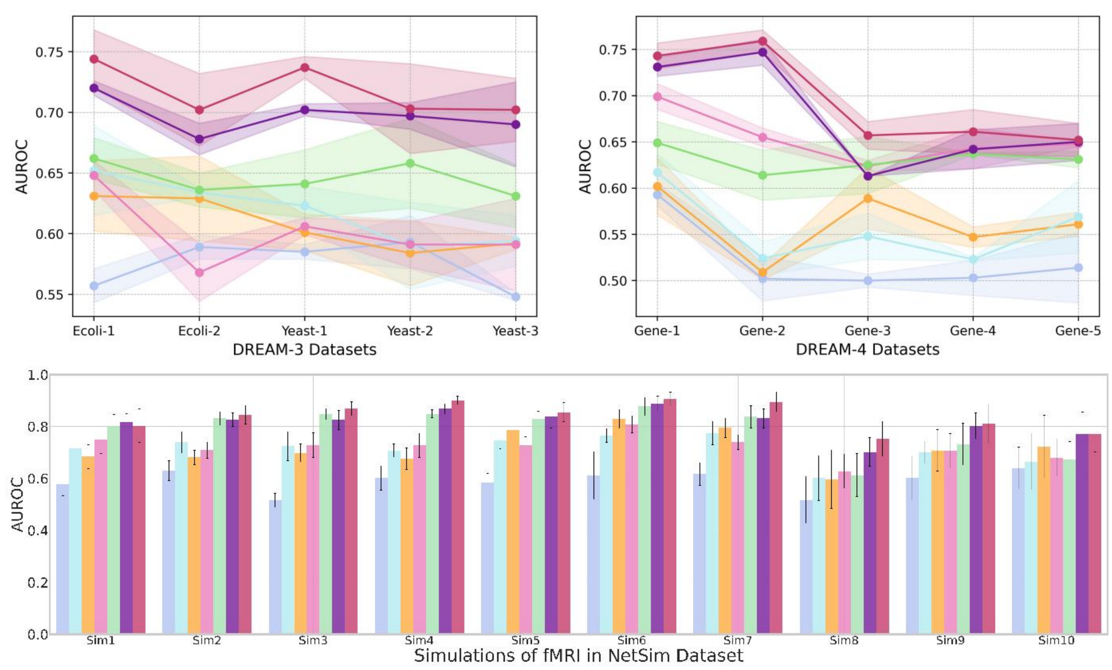
Synthetic Datasets

VAR, Lorenz–96

Synthetic Dataset	Metrics	GC	PCMCI	NGC	CR-VAE	CUTS	JRNGC	SpikGC
VAR(20,1000,5)	AUROC (↑)	0.598±0.020	0.666±0.020	0.925±0.015	0.947±0.010	0.833±0.025	0.970±0.019	**0.982±0.033**
	AUPRC (↑)	0.743±0.017	0.745±0.015	0.965±0.018	0.980±0.032	0.840±0.015	0.970±0.020	**0.989±0.018**
	F1 (↑)	0.701±0.028	0.733±0.015	0.930±0.024	0.963±0.015	0.852±0.017	0.969±0.017	**0.980±0.017**
	SHD (↓)	38±4	14±3	8±2	6±2	15±3	11±1	**5±1**
VAR(20,500,20)	AUROC (↑)	0.569±0.020	0.698±0.010	0.830±0.013	0.842±0.025	0.820±0.020	0.935±0.015	**0.971±0.012**
	AUPRC (↑)	0.588±0.025	0.675±0.025	0.835±0.015	0.837±0.027	0.810±0.021	0.925±0.016	**0.973±0.006**
	F1 (↑)	0.708±0.018	0.678±0.002	0.810±0.025	0.823±0.015	0.823±0.005	0.946±0.012	**0.955±0.002**
	SHD (↓)	179±4	165±12	99±5	22±3	65±15	59±7	**18±2**
VAR(40,1000,20)	AUROC (↑)	0.599±0.029	0.649±0.015	0.785±0.026	0.838±0.020	0.789±0.015	0.919±0.003	**0.945±0.005**
	AUPRC (↑)	0.578±0.020	0.643±0.017	0.845±0.015	0.837±0.017	0.813±0.021	0.902±0.018	**0.956±0.010**
	F1 (↑)	0.710±0.010	0.638±0.022	0.805±0.015	0.810±0.029	0.790±0.010	0.938±0.009	**0.950±0.002**
	SHD (↓)	164±3	85±2	83±10	79±6	58±9	33±6	**10±2**
Lorenz(20,1000,10)	AUROC (↑)	0.633±0.026	0.713±0.020	0.923±0.013	0.850±0.020	0.862±0.018	0.934±0.018	**0.979±0.033**
	AUPRC (↑)	0.610±0.014	0.715±0.028	0.893±0.020	0.867±0.021	0.875±0.009	0.946±0.015	**0.984±0.018**
	F1 (↑)	0.606±0.024	0.728±0.022	0.903±0.006	0.822±0.019	0.873±0.021	0.931±0.011	**0.980±0.017**
	SHD (↓)	48±2	29±3	9±1	14±3	12±2	10±2	**8±1**
Lorenz(20,500,20)	AUROC (↑)	0.540±0.018	0.656±0.023	0.853±0.020	0.813±0.038	0.775±0.016	0.903±0.020	**0.943±0.008**
	AUPRC (↑)	0.568±0.010	0.665±0.012	0.867±0.018	0.862±0.017	0.780±0.014	0.925±0.022	**0.950±0.015**
	F1 (↑)	0.690±0.018	0.725±0.013	0.565±0.017	0.810±0.026	0.770±0.010	0.915±0.004	**0.944±0.006**
	SHD (↓)	197±3	141±5	103±8	70±19	82±7	78±8	**23±3**
Lorenz(40,1000,20)	AUROC (↑)	0.560±0.019	0.716±0.018	0.743±0.021	0.825±0.006	0.719±0.015	0.907±0.008	**0.932±0.005**
	AUPRC (↑)	0.642±0.021	0.687±0.017	0.809±0.017	0.829±0.005	0.766±0.011	0.909±0.017	**0.940±0.016**
	F1 (↑)	0.571±0.015	0.755±0.010	0.768±0.024	0.774±0.017	0.767±0.029	0.913±0.002	**0.938±0.009**
	SHD (↓)	169±8	90±8	91±9	77±10	152±10	32±10	**28±5**

Real–world Datasets

fMRI, DREAM–3, DREAM–4



Effiency Performance

Model	Type	Tunable Parameters	MACs	Total time (s)
GC	ANN	10,516	25.6	180.5
PCMCI	-	-	36.5	174.8
NGC	ANN	104,210	31.4	207.3
CR-VAE	ANN	264,210	35.2	144.7
CUTS	ANN	286,640	30.5	142.7
JRNGC	ANN	33,210	22.5	133.6
SpikGC(ours)	SNN	30,808	13.6	30.8

Effiency Performance